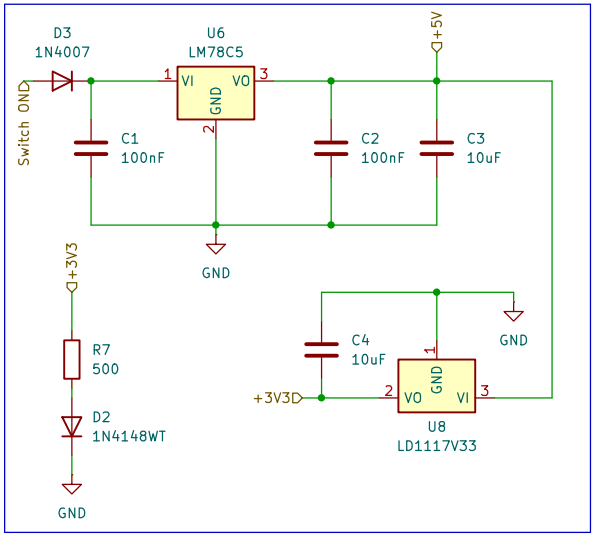
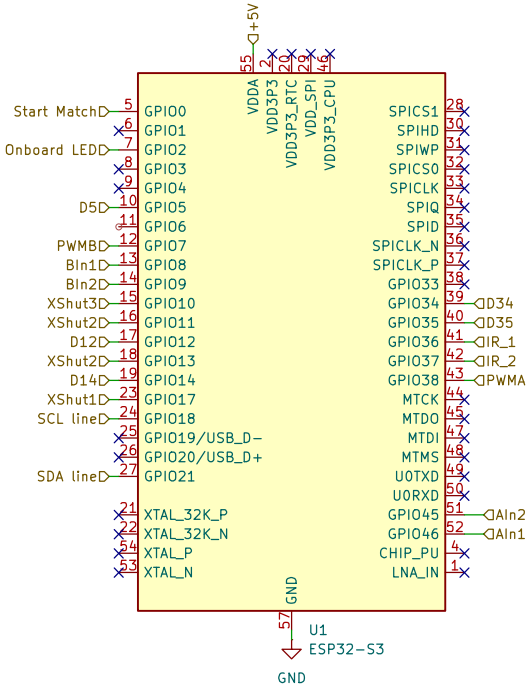
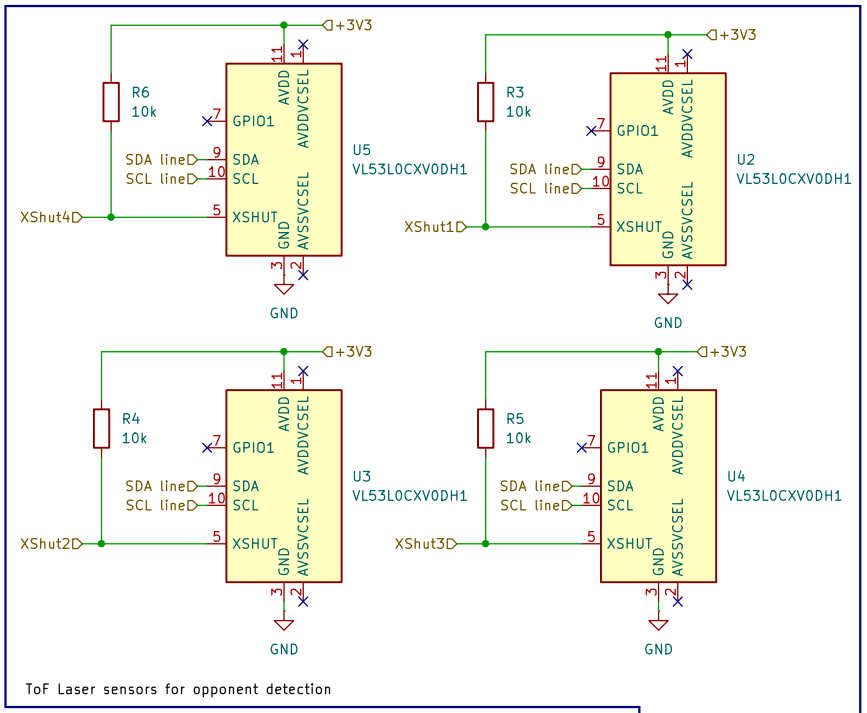
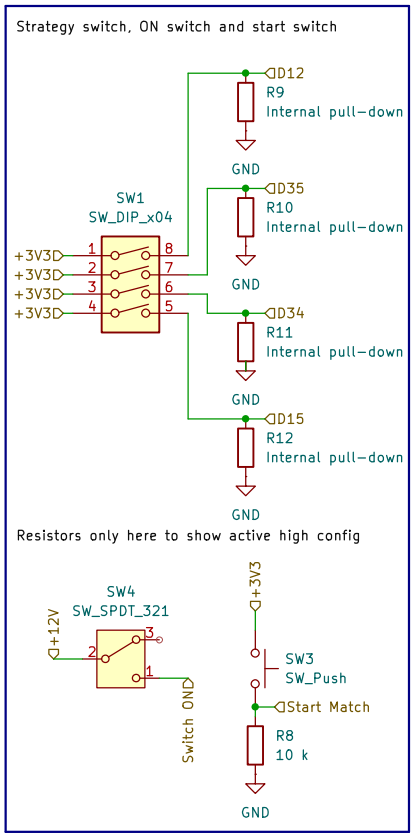
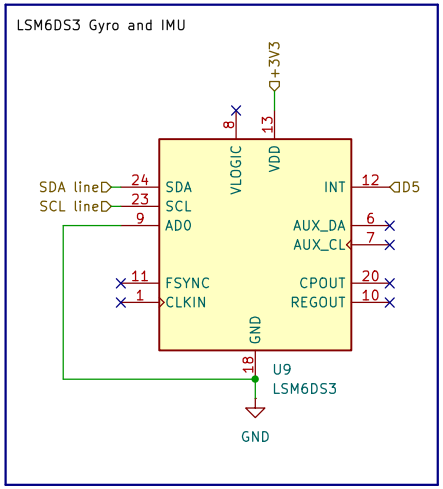
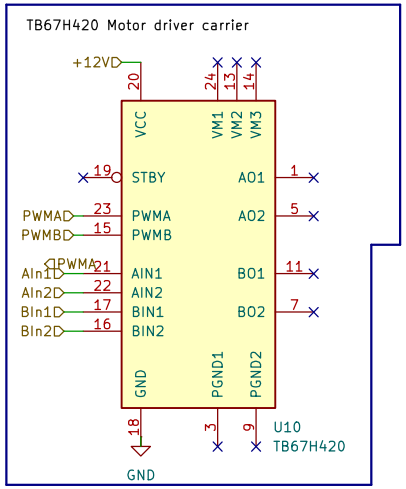

D3
1N4007
U6
LM78C5
+5V
Switch OND
VI
GND
VO
C1
100nF
C2
100nF
C3
10uF
GND
+3V3
R7
500
D2
1N4148WT
GND
C4
10uF
GND
+3V3
VO
GND
VI
U8
LD1117V33

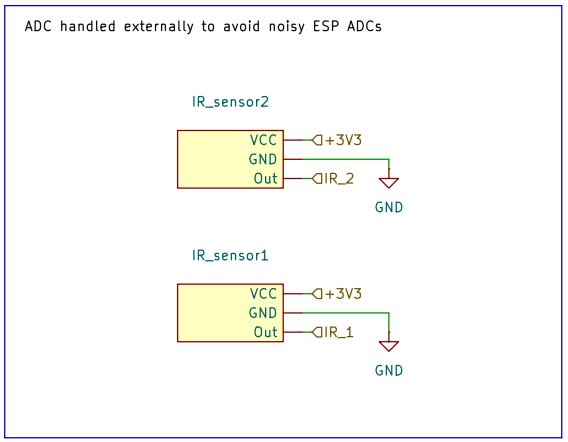

+5V
VDDA
VDD3P3
VDD3P3_RTC
VDD_SPI
VDD3P3_CPU
Start Match
GPIO0
Onboard LEDD
GPIO2
D5D
GPIO5
PWMBD
GPIO7
BIn1D
GPIO8
BIn2D
GPIO9
XShut3D
GPIO10
XShut2D
GPIO11
D12D
GPIO12
XShut2D
GPIO13
D14D
GPIO14
XShut1D
GPIO17
SCL lineD
GPIO18
GPIO19/USB_D−
GPIO20/USB_D+
SDA lineD
GPIO21
XTAL_32K_P
XTAL_32K_N
XTAL_P
XTAL_N
SPICS1
SPIHD
SPIWP
SPICS0
SPICLK
SPIQ
SPID
SPICLK_N
SPICLK_P
GPIO33
GPIO34
D34
GPIO35
D35
GPIO36
IR_1
GPIO37
IR_2
GPIO38
PWMA
MTCK
MTDO
MTDI
MTMS
U0TXD
U0RXD
GPIO45
AIn2
GPIO46
AIn1
CHIP_PU
LNA_IN
GND
U1
ESP32−S3
GND

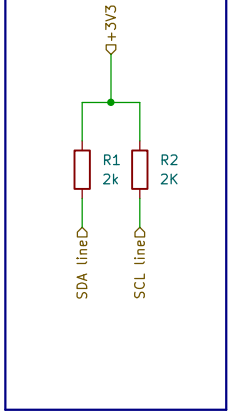

+3V3
R6
10k
GPIO1
SDA line
SCL line
SDA
SCL
XShut4
XSHUT
AVDD
AVDDVCSEL
AVSSVCSEL
GND
U5
VL53L0CXV0DH1
R3
10k
XShut1
U2
VL53L0CXV0DH1
R4
10k
XShut2
U3
VL53L0CXV0DH1
R5
10k
XShut3
U4
VL53L0CXV0DH1
ToF Laser sensors for opponent detection


Strategy switch, ON switch and start switch
SW1
SW_DIP_x04
+3V3
D12
R9
Internal pull−down
D35
R10
D34
R11
D15
R12
GND
Resistors only here to show active high config
SW4
SW_SPDT_321
+12V
Switch OND
SW3
SW_Push
Start Match
R8
10 k


LSM6DS3 Gyro and IMU
+3V3
VLOGIC
VDD
SDA line
SCL line
SDA
SCL
ADO
INT
D5
AUX_DA
AUX_CL
FSYNC
CLKIN
CPOUT
REGOUT
GND
U9
LSM6DS3
GND


TB67H420 Motor driver carrier
+12VD
VCC
VM1
VM2
VM3
STBY
PWMA
PWMB
AIN1
AIN2
BIN1
BIN2
AO1
AO2
BO1
BO2
PGND1
PGND2
GND
U10
TB67H420
GND


ADC handled externally to avoid noisy ESP ADCs
IR_sensor2
VCC
GND
Out
+3V3
IR_2
GND
IR_sensor1
VCC
GND
Out
+3V3
IR_1
GND


+3V3
R1
2k
R2
2K
SDA line
SCL line